SEQUENCE 3

# Straight lines in the plane

1–16 October 2026

## 3.1 Direction vector and Cartesian equation

**Key point · Definition.** A direction vector of a line is a **nonzero** vector having the direction of that line. If $A \neq B$, $\overrightarrow{AB}$ directs $(AB)$. Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

**Example 1.** For $A(-2, 4)$ and $B(6, 2)$, calculate $\overrightarrow{AB}$ then check that $\vec{u}(4, -1)$ directs $(AB)$.

**Key point · Establishing an equation with the determinant.** The line through $A(x_A, y_A)$ with direction vector $\vec{u}(\alpha, \beta) \neq \vec{0}$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

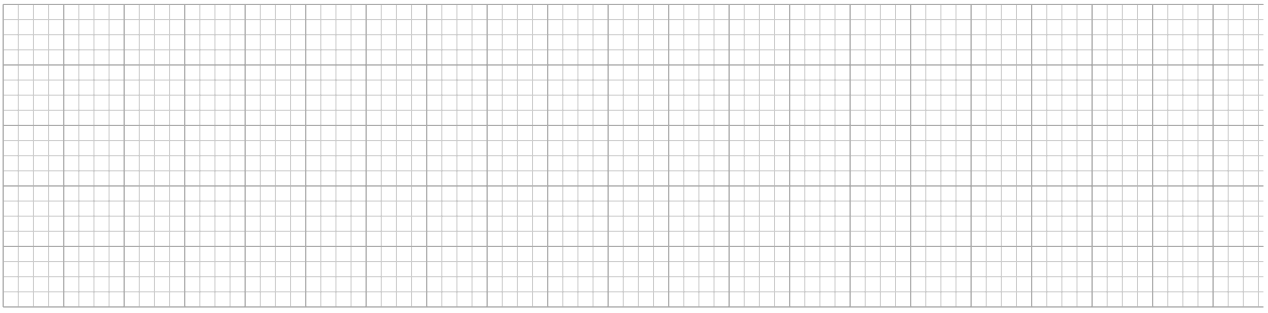
**Proof to write.**

**Watch out.** The zero vector directs no line. The sign of the determinant depends on the order of the vectors, but whether it vanishes does not.

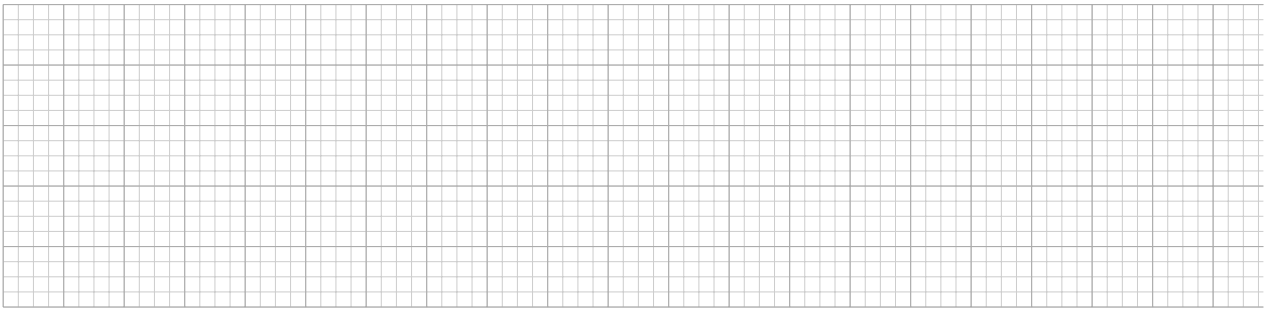
## 3.2 Determining the equation of a line

**Method.** From a point and a direction vector: choose $a, b$ then replace $x, y$ by the coordinates of the point to find $c$. From two distinct points: first calculate a direction vector.

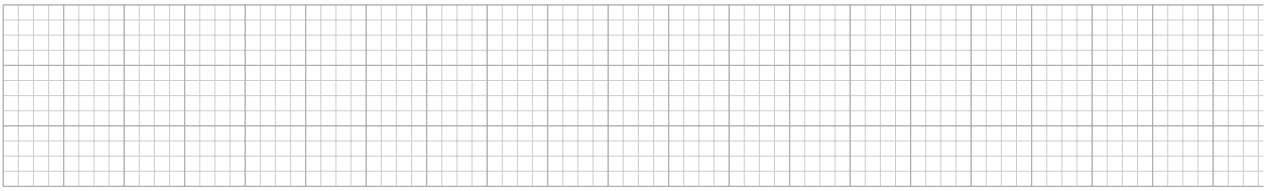
**Example 2.** Determine a Cartesian equation of the line through $A(2, -1)$ with direction vector $\vec{u}(3, 2)$.



**Example 3.** Determine a Cartesian equation of $(AB)$ for $A(3, 3)$ and $B(2, -2)$.



**Example 4.** Does the point $P(1, 2)$ belong to $d : 2x - 3y + 4 = 0$? Does the point $Q(2, 1)$ belong to it?

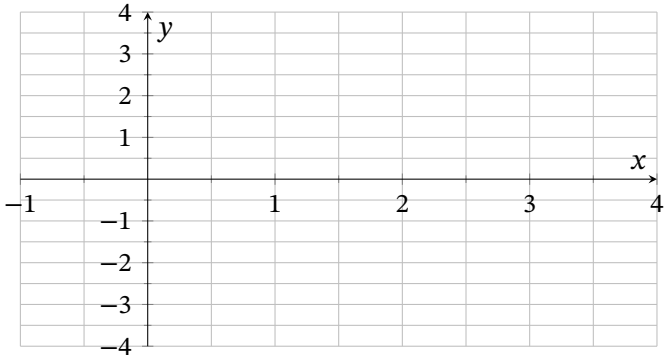


## 3.3 Slope-intercept equation, slope and affine function

**Key point · Definitions.** A non-vertical line has a unique slope-intercept equation $y = mx + p$. $m$ is the slope, $p$ the y-intercept. It represents the affine function $x \mapsto mx + p$; $(1, m)$ is a direction vector. A vertical line has an equation $x = k$; it is not the graph of a function of $x$. A horizontal line has an equation $y = k$.

**Key point · Calculating a slope.** If $x_A \neq x_B$, the slope of $(AB)$ is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find $p$, we then use $y_A = mx_A + p$.

**Example 5.** Determine the slope-intercept equation of $2x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.




**Example 6.** Find the affine function $f$ such that $f(0) = 3$ and $f(-1) = 5$, then the affine function $g$ whose graph passes through $C(-1, 2)$ and $E(5, -2)$.

## 3.4 Relative positions of two lines

**Key point · Properties.** Two lines are either intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$:

- if $m \neq m'$, they intersect;
- if $m = m'$ and $p \neq p'$, they are strictly parallel;
- if $m = m'$ and $p = p'$, they coincide.

For Cartesian equations, we compare the directions with the determinant. If the directions are collinear, we then test the membership of a point to distinguish coincident lines from strictly parallel ones.

**Example 7.** Determine an equation of $d$ through $A(2, 3)$ and parallel to $d' : x - 3y - 3 = 0$.

**Example 8.** Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$;
2. $x = 2$ and $x = -1$;
3. $y = 2x - 3$ and $y = -x + 3$.

## 3.5 Intersection: solving by substitution

**Key point · Definition.** Solving a system of two equations in $x, y$ means finding all the pairs $(x, y)$ satisfying both equations simultaneously. For two equations of lines: intersecting lines give one solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

**Method · Substitution.** Isolate one unknown in one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

**Example 9.** Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

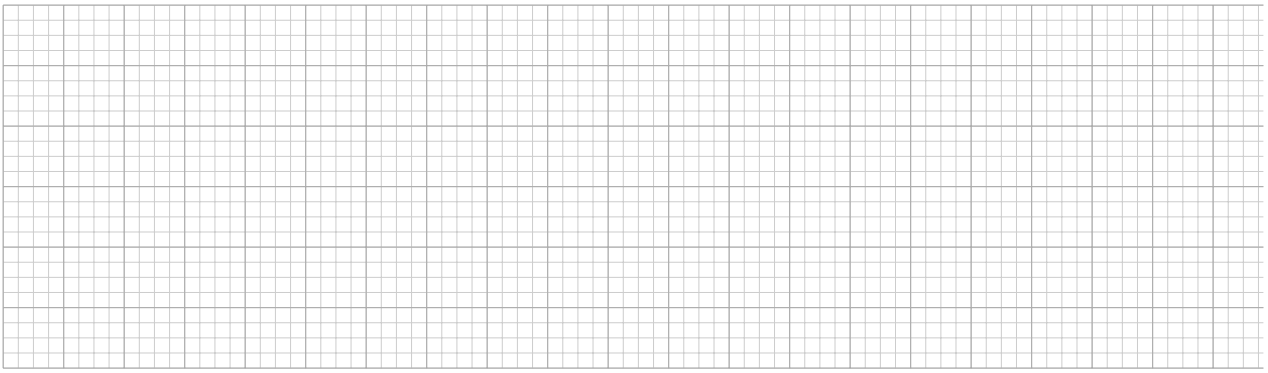
**Example 10.** Solve by substitution:

(a) $\begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases}$ (b) $\begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$

## 3.6 Linear combination and Python

**Method · Linear combination.** Multiply the equations by nonzero numbers to obtain opposite coefficients for one unknown; add them to eliminate that unknown. Keep one equation of the system to calculate the other unknown.

**Example 11.** Solve $\begin{cases} 2x+3y+1=0 \\ 3x-4y-2=0. \end{cases}$ Then redo the two systems of the previous example by linear combination.



```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

**Example 12.** Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

